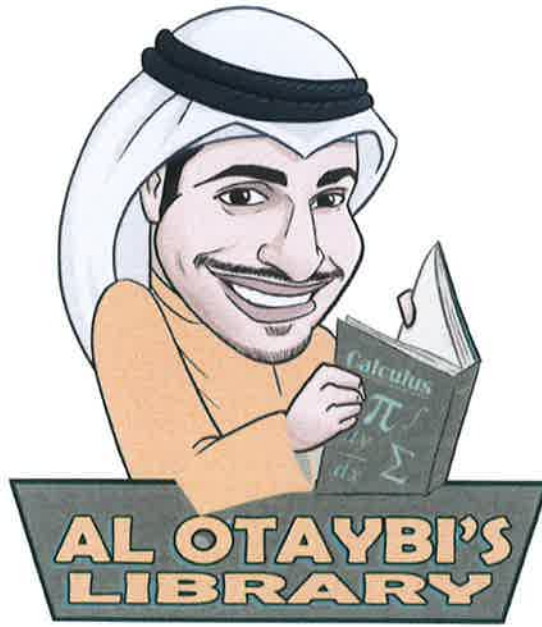


Kuwait University

2021/2022

Spring course



Calculus B

Final

مراجعة

اختبارات سابقة

(مراجعة الفايصل)

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Spring Semester 2019: Math 102 (Calculus II)
Final Exam

1. [4×5 = 20 pts.] In each of the following multiple choice questions, select the correct answer.

I. $\lim_{x \rightarrow 1} \frac{1 - e^{x-1}}{\sin(x-1)}$

- (A) does not exist (B) is 0 (C) is 1 (D) is -1 (E) None of these

II. $\int_0^{\pi} x \cos x \, dx =$

- (A) 1 (B) -2 (C) is 0 (D) is -1 (E) None of these

III. The sum of the series $\sum_{n=0}^{\infty} (-1)^n \binom{2/3}{n} \frac{1}{2^n}$ is

- (A) 2/3 (B) 1/2 (C) $1/\sqrt[3]{4}$ (D) $\sqrt[3]{4}$ (E) None of these

IV. Let $a_n = \frac{1 - 3n^2 + 2n^3}{5n^3 + 4}$, for $n = 1, 2, \dots$. Then,

- (A) $\{a_n\}_{n=1}^{+\infty}$ diverges and $\sum_{n=0}^{\infty} a_n$ converges (B) $\{a_n\}_{n=1}^{+\infty}$ converges and $\sum_{n=0}^{\infty} a_n$ diverges
(C) Both $\{a_n\}_{n=1}^{+\infty}$ and $\sum_{n=0}^{\infty} a_n$ diverge (D) Both $\{a_n\}_{n=1}^{+\infty}$ and $\sum_{n=0}^{\infty} a_n$ converge
(E) None of these

2. [10 pts.] Evaluate $\int \frac{x}{\sqrt{x^2 + 2x + 5}} dx$.

3. [5 + 5 = 10 pts.]

(a) Find the Maclaurin series of $f(x) = x^2 \tan^{-1}(x^3)$ and its radius of convergence.

(b) Find the radius and interval of convergence of the power series $\sum_{n=0}^{\infty} \frac{n^{10} 10^n}{n!} (x+1)^n$.

4. [10 pts.] Find the centroid of the region bounded by the curves $y = x^2$ and $x = y^3$.

5. [6 + 4 = 10 pts.] Let C be the curve with parametric equations: $x = t^3 - 2t$, $y = t^2 + 4$.

(a) Find the equation of the tangent line at the point $(-1, 5)$.

(b) Set-up (don't evaluate) an integral that represents the arc length of C corresponding to $1 \leq t \leq 2$.

6. [10 pts.] Let $\mathcal{R}$ be the region bounded by the curves $y^2 = x$ and $y^2 = 2 - x$.

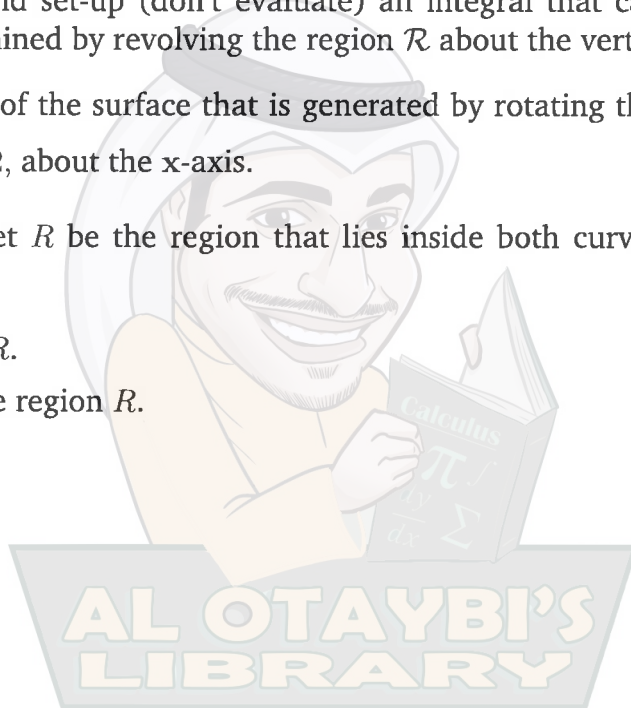
Sketch the region $\mathcal{R}$ and set-up (don't evaluate) an integral that can be used to find the volume of the solid obtained by revolving the region $\mathcal{R}$ about the vertical line $x = 3$.

7. [15 pts.] Find the area of the surface that is generated by rotating the portion of the curve $x = \frac{y^2}{4} - \frac{\ln y}{2}$, $1 \leq y \leq 2$, about the x-axis.

8. [5 + 10 = 15 pts.] Let R be the region that lies inside both curves: $r = 1 - \cos \theta$ and $r = -3 \cos \theta$.

(a) Sketch the region R .

(b) Find the area of the region R .





Multiple choices



$$\lim_{x \rightarrow 1} \frac{1 - e^{x-1}}{\sin(x-1)}$$

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(a) does not exist (b) is 0 (c) is 1 (d) is -1 (e) None of these

$$\lim_{x \rightarrow 1} \frac{1 - e^{x-1}}{\sin(x-1)} = \frac{1-1}{0} = \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 1} \frac{-e^{x-1}}{\cos(x-1)} = \frac{-1}{1} = -1$$

$$\therefore \lim_{x \rightarrow 1} \frac{1 - e^{x-1}}{\sin(x-1)} = -1 \quad (d)$$

$$\int_0^{\pi} x \cos x \, dx =$$

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(a) 1 (b) -2 (c) 0 (d) -1 (e) None of these

$$u = x \quad dv = \cos x \, dx$$

$$du = dx \quad v = \sin x$$

$$\left[x \sin x - \int \sin x \, dx \right]_0^{\pi}$$

$$\left[x \sin x + \cos x \right]_0^{\pi}$$

$$(0 - 1) - (0 + 1) = -1 - 1 = -2 \quad (b)$$



Multiple choices



The sum of the series $\sum_{n=0}^{\infty} (-1)^n \binom{2/3}{n} \frac{1}{2^n}$ is

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- (a) $2/3$ (b) $1/2$ (c) $1/\sqrt[3]{4}$ (d) $\sqrt[3]{4}$ (e) None of these

$$\sum_{n=0}^{\infty} (-1)^n \binom{\frac{2}{3}}{n} \left(\frac{1}{2}\right)^n$$

$$\sum_{n=0}^{\infty} \binom{\frac{2}{3}}{n} \left(-\frac{1}{2}\right)^n$$

$$u = -\frac{1}{2}$$

$$k = \frac{2}{3}$$

$$S = (1+u)^k$$

$$= \left(1 + \frac{-1}{2}\right)^{\frac{2}{3}} = \left(\frac{1}{2}\right)^{\frac{2}{3}} = \frac{1}{2^{\frac{2}{3}}}$$

$$= \frac{1}{4^{\frac{1}{3}}} = \frac{1}{\sqrt[3]{4}} \quad (c)$$



Multiple choices



Let $a_n = \frac{1 - 3n^2 + 2n^3}{5n^3 + 4}$, for $n = 1, 2, \dots$. Then

(A) $\{a_n\}_{n=1}^{+\infty}$ diverges and $\sum_{n=0}^{\infty} a_n$ converges.

(B) $\{a_n\}_{n=1}^{+\infty}$ converges and $\sum_{n=0}^{\infty} a_n$ diverges.

(c) Both $\{a_n\}_{n=1}^{+\infty}$ and $\sum_{n=0}^{\infty} a_n$ diverge.

(d) Both $\{a_n\}_{n=1}^{+\infty}$ and $\sum_{n=0}^{\infty} a_n$ converge.

(e) None of these

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$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1 - 3n^2 + 2n^3}{5n^3 + 4} = \lim_{n \rightarrow \infty} \frac{n^3 \left[\frac{1}{n^3} - \frac{3}{n} + 2 \right]}{n^3 \left[5 + \frac{4}{n^3} \right]} = \frac{0 - 0 + 2}{5 + 0} = \frac{2}{5}$$

$\therefore a_n$ is convergent

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} \frac{1 - 3n^2 + 2n^3}{5n^3 + 4}$$

$$\lim_{n \rightarrow \infty} \frac{1 - 3n^2 + 2n^3}{5n^3 + 4} = \frac{2}{5} \neq 0$$

$\therefore \sum_{n=0}^{\infty} a_n$ is Divergent

(test of divergence)

(B)

Evaluate $\int \frac{x}{\sqrt{x^2 + 2x + 5}} dx$

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$$\int \frac{x}{\sqrt{(x+1)^2 + 4}} dx$$

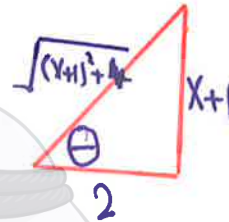
$$\begin{aligned} x^2 + 2x + 5 \\ (x+1)^2 - 1 + 5 \\ (x+1)^2 + 4 \end{aligned}$$

Let

$$x+1 = 2 \tan \theta \rightarrow \tan \theta = \frac{x+1}{2}$$

$$dx = 2 \sec^2 \theta d\theta$$

$$x = 2 \tan \theta - 1$$



$$\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\int \frac{2 \tan \theta - 1}{\sqrt{(2 \tan \theta)^2 + 4}} \cdot 2 \sec^2 \theta d\theta$$

$$\sqrt{4 \tan^2 \theta + 4} = \sqrt{4(\tan^2 \theta + 1)} = \sqrt{4 \sec^2 \theta} = 2 \sec \theta$$

$$\int \frac{2 \tan \theta - 1}{2 \sec \theta} \cdot 2 \sec^2 \theta d\theta$$

$$\int (2 \tan \theta - 1) \sec \theta d\theta$$

$$\int (2 \tan \theta \sec \theta - \sec \theta) d\theta$$

$$= 2 \sec \theta - \ln |\sec \theta + \tan \theta| + C$$

$$= 2 \frac{\sqrt{(x+1)^2 + 4}}{2} - \ln \left| \frac{\sqrt{(x+1)^2 + 4}}{2} + \frac{x+1}{2} \right| + C$$

(a) Find the Maclaurin series of $f(x) = x^2 \tan^{-1}(x^3)$ and its radius of convergence.

(b) Find the radius and interval of convergence of the power series

$$\sum_{n=0}^{\infty} \frac{n^{10} 10^n}{n!} (x+1)^n$$

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$$\begin{aligned} \text{(a)} \quad x^2 \tan^{-1}(x^3) &= x^2 \sum_{n=0}^{\infty} (-1)^n \frac{(x^3)^{2n+1}}{2n+1} = x^2 \sum_{n=0}^{\infty} (-1)^n \frac{x^{6n+3}}{2n+1} \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{6n+5}}{2n+1} \end{aligned}$$

$|x^3| < 1 \rightarrow |x| < 1$
 $R = 1$

$$\begin{aligned} \text{(b)} \quad U_n &= \frac{n^{10} 10^n}{n!} (x+1)^n \rightarrow U_{n+1} = \frac{(n+1)^{10} 10^{n+1}}{(n+1)!} (x+1)^{n+1} \\ \lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(n+1)^{10} 10^{n+1}}{(n+1) n!} (x+1)^{n+1} (x+1)^{-n} \frac{n!}{n^{10} 10^n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{(n+1)^9}{n^{10}} 10(x+1) \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{[n(1+\frac{1}{n})]^9}{n^{10}} 10(x+1) \right| = \lim_{n \rightarrow \infty} \left| \frac{n^9 (1+\frac{1}{n})^9}{n^{10}} 10(x+1) \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{(1+\frac{1}{n})^9}{n} 10(x+1) \right| = 0 < 1 \end{aligned}$$

$\therefore$ The interval of convergence = $(-\infty, \infty)$

The radius of convergence $R = \infty$

Find the centroid of the region bounded by the curves

$$y = x^2 \text{ and } x = y^3.$$

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$$y_1 = x^2 \quad y_2 = x^{\frac{1}{3}}$$

$$y = y$$

$$x^2 = x^{\frac{1}{3}}$$

$$x^6 = x$$

$$x^6 - x = 0 \rightarrow x(x^5 - 1) = 0 \rightarrow \boxed{x=0} \quad \boxed{x=1}$$

at $x = \frac{1}{2} \rightarrow \begin{matrix} y_1 = \frac{1}{4} \\ y_2 = \frac{1}{2^{\frac{1}{3}}} \end{matrix} \rightarrow \therefore y_2 > y_1$

$$A = \int_0^1 [x^{\frac{1}{3}} - x^2] dx = \left[\frac{3}{4} x^{\frac{4}{3}} - \frac{x^3}{3} \right]_0^1 = \frac{3}{4} - \frac{1}{3} = \frac{5}{12}$$

$$\begin{aligned} \bar{X} &= \frac{1}{A} \int_0^1 x [x^{\frac{1}{3}} - x^2] dx \\ &= \frac{12}{5} \int_0^1 [x^{\frac{4}{3}} - x^3] dx = \frac{12}{5} \left[\frac{3}{7} x^{\frac{7}{3}} - \frac{x^4}{4} \right]_0^1 \\ &= \frac{12}{5} \left[\frac{3}{7} - \frac{1}{4} \right] = \frac{12}{5} \cdot \frac{12-7}{28} = \frac{12}{5} \cdot \frac{5}{28} = \frac{3}{7} \end{aligned}$$

$$\begin{aligned} \bar{Y} &= \frac{1}{2A} \int_0^1 [(x^{\frac{1}{3}})^2 - (x^2)^2] dx \\ &= \frac{1}{2} \cdot \frac{12}{5} \int_0^1 [x^{\frac{2}{3}} - x^4] dx = \frac{6}{5} \left[\frac{3}{5} x^{\frac{5}{3}} - \frac{x^5}{5} \right]_0^1 = \frac{6}{5} \left[\frac{3}{5} - \frac{1}{5} \right] = \frac{6}{5} \cdot \frac{2}{5} = \frac{12}{25} \end{aligned}$$

$$\Rightarrow \therefore P(\bar{X}, \bar{Y}) = \left(\frac{3}{7}, \frac{12}{25} \right)$$

Let C be the curve with parametric equations :

$$x = t^3 - 2t, y = t^2 + 4$$

a) Find the equation of the tangent line at the point $(-1, 5)$.

b) Set-up (don't evaluate) an integral that represents the arc length of C corresponding to $1 \leq t \leq 2$.

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(a)

$$\text{at } (-1, 5) \begin{cases} -1 = t^3 - 2t & \textcircled{1} \\ 5 = t^2 + 4 \end{cases}$$

$$t^2 = 1 \begin{cases} t = -1 \\ t = 1 \end{cases}$$

$$\text{in } \textcircled{1} \rightarrow -1 = -1 + 2 \rightarrow -1 = 1 \text{ rejected}$$

$$\text{in } \textcircled{1} \rightarrow -1 = 1 - 2 \rightarrow -1 = -1 \text{ OK}$$

$$\therefore \boxed{t = 1}$$

$$\frac{dx}{dt} = 3t^2 - 2 \rightarrow \text{at } t = 1 \rightarrow \boxed{\frac{dx}{dt}} = 3(1) - 2 = \boxed{1}$$

$$\frac{dy}{dt} = 2t \rightarrow \text{at } t = 1 \rightarrow \boxed{\frac{dy}{dt}} = 2(1) = \boxed{2}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2}{1} = 2$$

$\therefore$ The equation of the tangent line:

$$y - y_0 = m(x - x_0)$$

$$y - 5 = 2(x - (-1))$$

$$(b) L = \int_1^2 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = 3t^2 - 2 \rightarrow \left(\frac{dx}{dt}\right)^2 = (3t^2 - 2)^2$$

$$\frac{dy}{dt} = 2t \rightarrow \left(\frac{dy}{dt}\right)^2 = (2t)^2$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (3t^2 - 2)^2 + (2t)^2$$

$$L = \int_1^2 \sqrt{(3t^2 - 2)^2 + (2t)^2} dt$$

Let R be the region bounded by the curves $y^2 = x$ and $y^2 = 2 - x$.
 Sketch the region R and set-up (don't evaluate) an integral that
 can be used to find the volume of the solid obtained by revolving
 the region R about the vertical line $x = 3$.

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$$y^2 = x \quad y^2 = 2 - x \quad \text{about } x = 3$$

$$x = 2 - y^2$$

$$x = x$$

$$y^2 = 2 - y^2$$

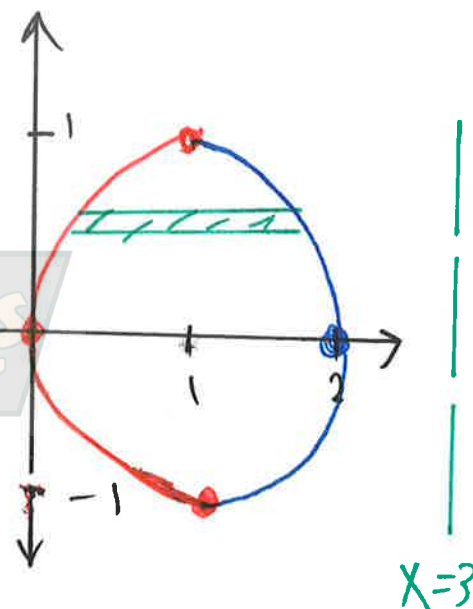
$$2y^2 = 2 \rightarrow y^2 = 1 \rightarrow y = -1 \text{ or } y = 1$$

$$x = y^2$$

x	1	0	1
y	-1	0	1

$$x = 2 - y^2$$

x	1	2	1
y	-1	0	1



$$V = \pi \int_{-1}^1 [(OR)^2 - (IR)^2] dy$$

$$= \pi \int_{-1}^1 [(3 - y^2)^2 - (3 - (2 - y^2))^2] dy$$

Find the area of the surface that is generated by rotating the portion of the curve

$$x = \frac{y^2}{4} - \frac{\ln y}{2}, \quad 1 \leq y \leq 2, \quad \text{about the } x\text{-axis}$$

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$$S = 2\pi \int_1^2 y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$\frac{dx}{dy} = \frac{2y}{4} - \frac{1}{2} \frac{1}{y} = \frac{y}{2} - \frac{1}{2y} = \frac{y^2}{2y} - \frac{1}{2y} = \frac{y^2 - 1}{2y}$$

$$\left(\frac{dx}{dy}\right)^2 = \left(\frac{y^2 - 1}{2y}\right)^2 = \frac{(y^2 - 1)^2}{(2y)^2} = \frac{y^4 - 2y^2 + 1}{4y^2}$$

$$\Rightarrow 1 + \left(\frac{dx}{dy}\right)^2 = 1 + \frac{y^4 - 2y^2 + 1}{4y^2} = \frac{4y^2}{4y^2} + \frac{y^4 - 2y^2 + 1}{4y^2} = \frac{y^4 + 2y^2 + 1}{4y^2}$$

$$S = 2\pi \int_1^2 y \sqrt{\frac{y^4 + 2y^2 + 1}{4y^2}} dy = 2\pi \int_1^2 y \frac{\sqrt{y^4 + 2y^2 + 1}}{\sqrt{4y^2}} dy$$

$$= \cancel{2\pi} \int_1^2 \cancel{y} \frac{\sqrt{y^4 + 2y^2 + 1}}{\cancel{2y}} dy = \frac{y^4 + 2y^2 + 1}{(y^2 + 1)^2 - 1 + 1}$$

$$= \pi \int_1^2 \sqrt{(y^2 + 1)^2} dy$$

$$= \pi \left[\frac{y^3}{3} + y \right]_1^2 = \pi \left[\left(\frac{8}{3} + 2 \right) - \left(\frac{1}{3} + 1 \right) \right]$$

$$= \pi \left[\frac{8}{3} + 2 - \frac{1}{3} - 1 \right] = \pi \left[\frac{7}{3} + 1 \right]$$

Let R be the region that lies inside both curves :

$$r = 1 - \cos\theta \text{ and } r = -3\cos\theta$$

(a) Sketch the region R .

(b) Find the area of the region R .

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$$r = 1 - \cos\theta \quad r = -3\cos\theta$$

$$r = r$$

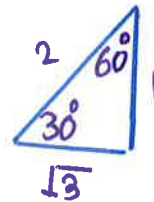
$$1 - \cos\theta = -3\cos\theta$$

$$1 = -2\cos\theta$$

$$\cos\theta = -\frac{1}{2}$$

$$\theta = \pi - \frac{\pi}{3} \rightarrow \theta = \frac{2\pi}{3}$$

$$\theta = \pi + \frac{\pi}{3} \rightarrow \theta = \frac{4\pi}{3}$$

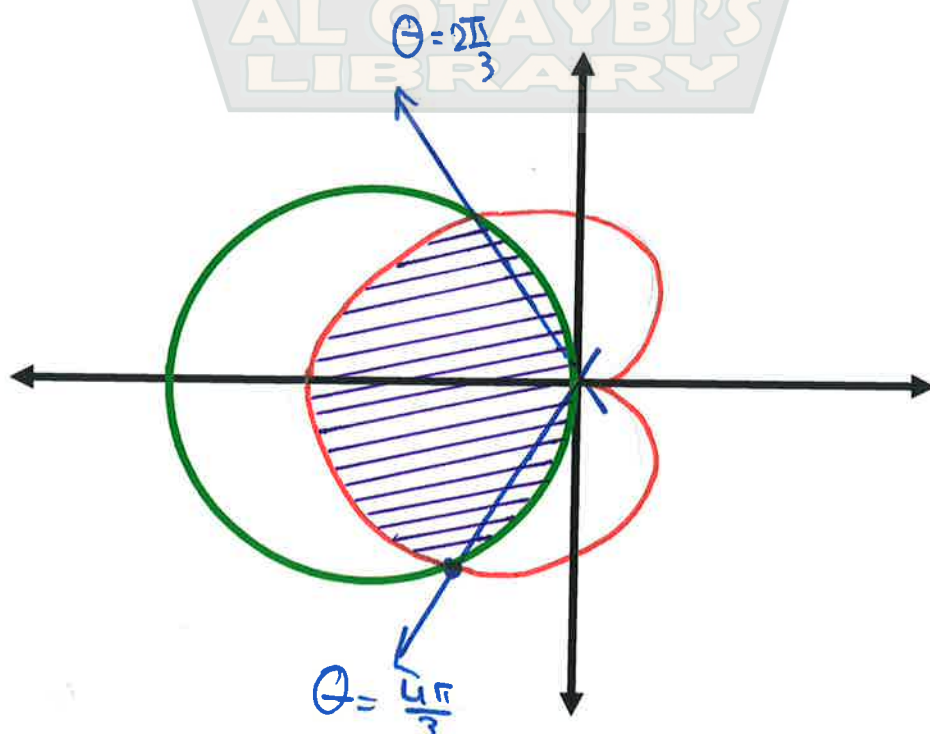


$$\frac{\pi}{3}$$

$$\pi - \alpha$$

$$\pi + \alpha$$

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$$A_1 = \frac{1}{2} \int_{\frac{2\pi}{3}}^{\pi} r^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{2\pi}{3}}^{\pi} (1 - \cos\theta)^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{2\pi}{3}}^{\pi} (1 - 2\cos\theta + \cos^2\theta) d\theta$$

$$= \frac{1}{2} \left[\theta - 2\sin\theta + \frac{1}{2}(\theta + \frac{\sin 2\theta}{2}) \right]_{\frac{2\pi}{3}}^{\pi}$$

$$= \frac{1}{2} \left[(\pi - 0) + \frac{1}{2}(\pi + 0) \right.$$

$$\left. - \left(\frac{2\pi}{3} - 2\frac{\sqrt{3}}{2} + \frac{1}{4} \left(\frac{2\pi}{3} + \frac{\sqrt{3}}{2} \right) \right) \right]$$

$$A_{\pi} = 2(A_1 + A_2)$$

$$= \pi + \frac{1}{2}\pi - \left(\frac{2\pi}{3} - \sqrt{3} + \frac{1}{2} \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{4} \right) \right)$$

$$+ \frac{9}{2} \left[\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{4} \right) - \frac{\pi}{2} \right]$$

$$A_{\pi} = \frac{5\pi}{4}$$

$$A_2 = \frac{1}{2} \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} r^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} (-3\cos\theta)^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} 9\cos^2\theta d\theta$$

$$= \frac{9}{2} \cdot \frac{1}{2} \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} (1 + \cos 2\theta) d\theta$$

$$= \frac{9}{4} \left[\theta + \frac{\sin 2\theta}{2} \right]_{\frac{\pi}{2}}^{\frac{2\pi}{3}}$$

$$= \frac{9}{4} \left[\left(\frac{2\pi}{3} + \frac{\sqrt{3}}{2} \right) - \left(\frac{\pi}{2} + 0 \right) \right]$$

Final 2017/2018

1. (5 pts.) Find $\lim_{x \rightarrow 0^+} \frac{\ln x}{\ln(\tan x)}$.

2. (5 pts.) Determine whether the improper integral $\int_0^{\pi/2} \tan x \, dx$ converges or diverges.

3. (5 pts.) Find the slope of the tangent line to the parametric curve:

$$x(t) = e^{t-1}, \quad y(t) = \tan^{-1} t$$

at the point where $t = 1$.

4. (5 pts.) Determine whether the sequence $\{a_n\}$ is convergent or divergent, where

$$a_n = \frac{2^n}{\sin(2n) + 3^n}.$$

5. (5 pts.) Find the sum of the series $\sum_{n=1}^{\infty} 2^{3n} 3^{1-2n}$.

6. (5 pts.) Find a polar equation $r = f(\theta)$ for the curve represented by Cartesian equation

$$x^2 + y^2 = 10x + 6y.$$

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مع السؤال الأخير

7. (10 pts.) Let $f(x) = x^2 \tan^{-1} x$. Find the Maclaurin series for $f'(x)$ and its radius of convergence.

8. (10 pts.) Find the volume of the solid obtained by rotating the region bounded by $y = e^x$ and $y = 0$ for $0 \leq x \leq 1$, about the line $y = -1$.

9. (10 pts.) Let C be the curve with parametric equations

$$x(t) = \frac{1}{2}t^2, \quad y(t) = \frac{1}{3}(2t + 1)^{3/2}, \quad 0 \leq t \leq 4.$$

Find its arc length.

10. (10 pts.) Find the area of the region under the parametric curve:

$$x(t) = t^4, \quad y(t) = \sqrt{t^2 + 1}, \quad 0 \leq t \leq 1.$$

11. (15 pts.) Find the centroid of the region bounded by $y = \sin(2x)$ and $y = 0$ for $0 \leq x \leq \frac{\pi}{2}$.

12. (5 + 10 pts.) Let $C_1 : r = 2 \cos \theta$ and $C_2 : r = \sqrt{3}$ be polar curves.

a) Write Cartesian equations for C_1 and C_2 .

b) Find the area inside C_1 and outside C_2 .



Find

Fall 2017

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{\ln(\tan x)} \quad \frac{\ln 0}{\ln 0} = \frac{-\infty}{-\infty}$$

L'Hop

$$\lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{\frac{\sec^2 x}{\tan x}}$$

$$\lim_{x \rightarrow 0^+} \frac{\tan x}{\sec^2 x \cdot x}$$

$$\lim_{x \rightarrow 0^+} \frac{\frac{\sin x}{\cos x}}{\frac{1}{\cos^2 x} \cdot x}$$

$$\lim_{x \rightarrow 0^+} \frac{\sin x \cos x}{x} \quad \frac{0}{0}$$

الدالة

$$\lim_{x \rightarrow 0^+} \frac{\tan x}{\sec^2 x \cdot x} = \frac{1}{(1)^2} = 1$$

L'Hop

$$\lim_{x \rightarrow 0^+} \frac{\cos x \cos x + \sin x (-\sin x)}{1} = \frac{(1)(1) + 0}{1} = 1$$

$$\therefore \lim_{x \rightarrow 0^+} \frac{\ln x}{\ln(\tan x)} = 1$$

Determine whether the improper integral

$$\int_0^{\pi/2} \tan x \, dx$$

converges or diverges.

Fall 2017

$$\frac{\sin x}{\cos x}$$

$$\text{الخارج} \\ x = \frac{\pi}{2}$$

$$I_1 = \int \tan x \, dx = \ln |\sec x| + C$$

$$\lim_{t \rightarrow \frac{\pi}{2}^-} \int_0^t \tan x \, dx$$

$$\lim_{t \rightarrow \frac{\pi}{2}^-} [\ln |\sec x|]_0^t$$

$$\lim_{t \rightarrow \frac{\pi}{2}^-} [\ln |\sec t| - \ln |\sec 0|]$$

$\frac{1}{\cos t}$

$$= \ln |\infty| - \ln |1|$$

$$= \infty - 0 = \infty$$

$$\therefore \int_0^{\pi/2} \tan x \, dx \text{ is Divergent.}$$

Find the slope of the tangent line to the parametric curve :

$$x(t) = e^{t-1}, \quad y(t) = \tan^{-1}t$$

at the point where $t = 1$.

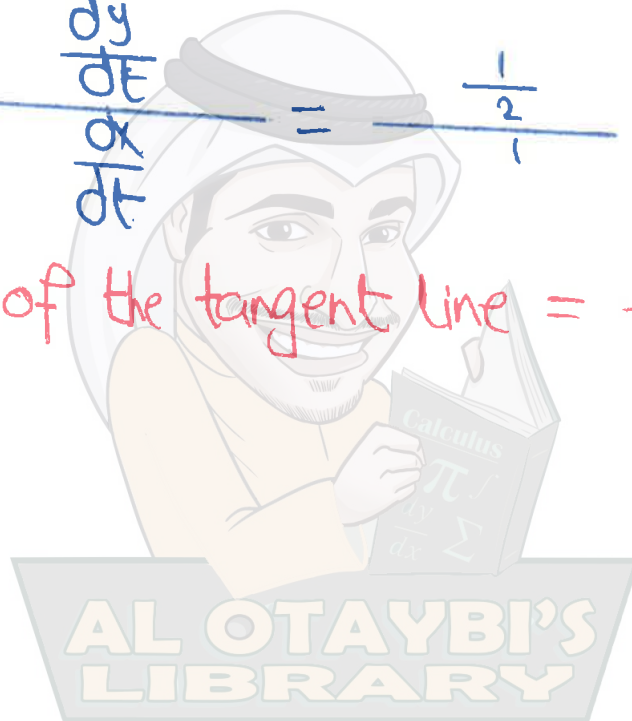
Fall 2017

$$\frac{dx}{dt} = e^{t-1} \xrightarrow{\text{at } t=1} \frac{dx}{dt} = e^0 = 1$$

$$\frac{dy}{dt} = \frac{1}{t^2+1} \xrightarrow{\text{at } t=1} \frac{dy}{dt} = \frac{1}{2}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{1}{2}}{1} = \frac{1}{2}$$

∴ The slope of the tangent line = $\frac{1}{2}$



Determine whether the sequence $\{a_n\}$ is convergent or divergent, where

$$a_n = \frac{2^n}{\sin(2n) + 3^n}$$

Fall 2017

$$\lim_{n \rightarrow \infty} \frac{2^n}{\sin(2n) + 3^n}$$

$$\div 3^n$$

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{2}{3}\right)^n}{\frac{\sin(2n)}{3^n} + 1} = \frac{0}{0 + 1} = 0$$

$$\lim_{n \rightarrow \infty} \frac{\sin(2n)}{3^n}$$

$$-1 \leq \sin(2n) \leq 1$$

$$\div 3^n$$

$$-\left(\frac{1}{3}\right)^n \leq \frac{\sin(2n)}{3^n} \leq \left(\frac{1}{3}\right)^n$$

$$\lim_{n \rightarrow \infty} -\left(\frac{1}{3}\right)^n = 0$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{3}\right)^n = 0$$

∴ By Sandwich Theorem

$$\lim_{n \rightarrow \infty} \frac{\sin(2n)}{3^n} = 0$$

∴ a_n is convergent

Find the sum of the series

$$\sum_{n=1}^{\infty} 2^{3n} 3^{1-2n} \rightarrow a_n$$

Fall 2017

$n=1$ $a_1 = 2^3 3^{1-2} = 2^3 3^{-1} = \frac{2^3}{3}$

$n=2$ $a_2 = 2^6 3^{1-4} = 2^6 3^{-3} = \frac{2^6}{3^3}$

$$r = \frac{a_2}{a_1} = \frac{\frac{2^6}{3^3}}{\frac{2^3}{3}} = \frac{2^3}{3^2} = \frac{8}{9}$$

$\therefore |r| < 1$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (Geometric Series)

$$\begin{aligned} S &= \frac{a}{1-r} \\ &= \frac{\frac{2^3}{3}}{1 - \frac{8}{9}} = \frac{\frac{8}{3}}{\frac{9}{9} - \frac{8}{9}} \\ &= \frac{\frac{8}{3}}{\frac{1}{9}} = 24 \end{aligned}$$

Find a polar equation $r = f(\theta)$ for the curve represented by Cartesian equation

$$x^2 + y^2 = 10x + 6y$$

Fall 2017

$$r^2 = 10r\cos\theta + 6r\sin\theta$$

(Let $r \neq 0$) $\div r$

$$r = 10\cos\theta + 6\sin\theta$$

Let $f(x) = x^2 \tan^{-1} x$. Find the Maclaurin series for $f'(x)$ and its radius of convergence.

Fall 2017

$$f(x) = x^2 \tan^{-1} x = x^2 \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

$$\therefore f(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+3}}{2n+1}$$

$$\therefore f'(x) = \sum_{n=0}^{\infty} (-1)^n \frac{(2n+3) x^{2n+2}}{2n+1}$$

Radius of convergence

$$|x| < 1$$

$$R = 1$$

Find the volume of the solid obtained by rotating the region bounded by $y = e^x$ and $y = 0$ for $0 \leq x \leq 1$, about the line $y = -1$.

Fall 2017

$$y = e^x \quad y = 0 \quad 0 \leq x \leq 1 \quad \text{about } y = -1$$

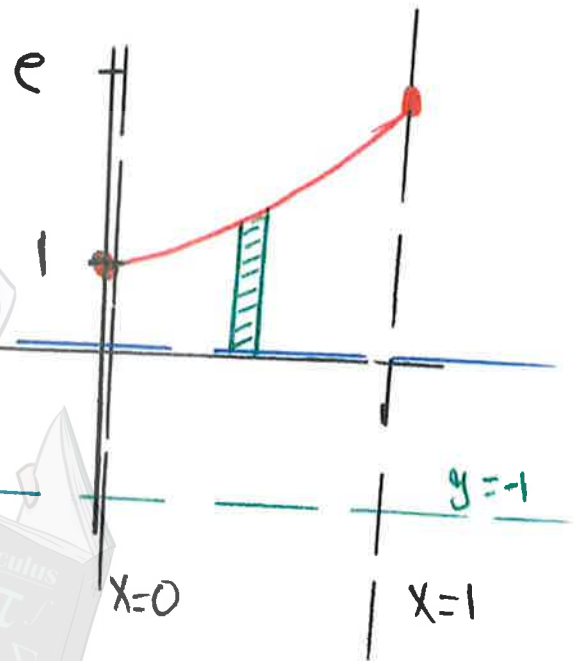
$$y = y$$

$$e^x = 0$$

None

$$y = e^x$$

x	0		1
y	1		e



$$\begin{aligned}
 V &= \pi \int_0^1 [(OR)^2 - (IR)^2] dx \\
 &= \pi \int_0^1 [(e^x - (-1))^2 - (0 - (-1))^2] dx \\
 &= \pi \int_0^1 [(e^x + 1)^2 - 1] dx \\
 &= \pi \int_0^1 [e^{2x} + 2e^x + 1 - 1] dx \\
 &= \pi \left[\frac{e^{2x}}{2} + 2e^x \right]_0^1 \\
 &= \pi \left[\frac{e^2}{2} + 2e - \left(\frac{1}{2} + 2 \right) \right]
 \end{aligned}$$

Let C be the curve with parametric equations

$$x(t) = \frac{1}{2}t^2, \quad y(t) = \frac{1}{3}(2t+1)^{3/2}, \quad 0 \leq t \leq 4.$$

Find the arc length.

Fall 2017

$$L = \int_0^4 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = t \longrightarrow \left(\frac{dx}{dt}\right)^2 = t^2$$

$$\frac{dy}{dt} = \frac{1}{3} \cdot \frac{3}{2} (2t+1)^{-1/2} \cdot 2 \longrightarrow \left(\frac{dy}{dt}\right)^2 = 2t+1$$

$$\longrightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = t^2 + 2t + 1$$

$$L = \int_0^4 \sqrt{t^2 + 2t + 1} dt$$

$$\begin{aligned} t^2 + 2t + 1 \\ (t+1)^2 - 1 + 1 \\ (t+1)^2 \end{aligned}$$

$$= \int_0^4 \sqrt{(t+1)^2} dt$$

$$= \int_0^4 (t+1) dt$$

$$= \left[\frac{t^2}{2} + t \right]_0^4$$

$$= \frac{16}{2} + 4 - 0$$

$$= 8 + 4 = 12$$

Find the area of the region under the parametric curve

$$x(t) = t^4, \quad y(t) = \sqrt{t^2 + 1}, \quad 0 \leq t \leq 1.$$

Fall 2017

$$A = \int_0^1 y \, dx$$

$$= \int_0^1 \sqrt{t^2 + 1} \cdot 4t^3 \, dt$$

$$= 4 \int_0^1 t^3 \sqrt{t^2 + 1} \, dt$$

$$\text{Let } y = \sqrt{t^2 + 1}$$

$$y^2 = t^2 + 1 \longrightarrow t^2 = y^2 - 1$$

$$2y = 2t \, dt$$

$$y = t \, dt$$

$$A = 4 \int_0^1 \boxed{t} \boxed{t^2} \boxed{\sqrt{t^2 + 1}} \boxed{dt}$$

$$= 4 \int (y^2 - 1) y \cdot y \, dy$$

$$= 4 \int (y^2 - 1) y^2 \, dy$$

$$= 4 \int (y^4 - y^2) \, dy$$

$$= 4 \left[\frac{y^5}{5} - \frac{y^3}{3} \right] = 4 \left[\frac{(\sqrt{t^2 + 1})^5}{5} - \frac{(\sqrt{t^2 + 1})^3}{3} \right] \Big|_0^1$$

$$= 4 \left[\left[\frac{(\sqrt{2})^5}{5} - \frac{(\sqrt{2})^3}{3} \right] - \left[\frac{1}{5} - \frac{1}{3} \right] \right]$$

Find the centroid of the region bounded by

Fall 2017

$$y = \sin(2x) \text{ and } y = 0 \text{ for } 0 \leq x \leq \frac{\pi}{2}$$

$$A = \int_0^{\frac{\pi}{2}} [\sin(2x) - 0] dx = \left[\frac{-\cos(2x)}{2} \right]_0^{\frac{\pi}{2}} = -\frac{\cos(\pi)}{2} + \frac{1}{2} = \frac{1}{2} + \frac{1}{2} = \boxed{1}$$

$$\bar{x} = \frac{1}{A} \int_0^{\frac{\pi}{2}} x [\sin(2x) - 0] dx$$

① $u = x \quad dv = \sin(2x)$
 $du = dx \quad v = \frac{-\cos(2x)}{2}$

$$= \left[\frac{-1}{2} x \cos(2x) \right]_0^{\frac{\pi}{2}} + \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos 2x dx$$

$$= \left[\frac{-1}{2} x \cos(2x) + \frac{1}{4} \sin(2x) \right]_0^{\frac{\pi}{2}}$$

$$= \left(\frac{-1}{2} \cdot \frac{\pi}{2} \cdot (-1) + 0 \right) - (0) = \boxed{\frac{\pi}{4}}$$

$$\bar{y} = \frac{1}{2A} \int_0^{\frac{\pi}{2}} [\sin^2(2x) - 0^2] dx$$

$$= \frac{1}{2(1)} \cdot \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 - \cos(4x)) dx$$

$$= \frac{1}{4} \left[x - \frac{\sin(4x)}{4} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{4} \left[\left(\frac{\pi}{2} - 0 \right) - 0 \right] = \boxed{\frac{\pi}{8}}$$

$$\Rightarrow P(\bar{x}, \bar{y}) = P\left(\frac{\pi}{4}, \frac{\pi}{8}\right)$$

Let $C_1: r = 2\cos\theta$ and $C_2: r = \sqrt{3}$ be polar curves.

Write the Cartesian equations for C_1 and C_2 .

Fall 2017

$$r = 2\cos\theta$$

*r

$$r^2 = 2r\cos\theta$$

$$x^2 + y^2 = 2x$$

$$x^2 - 2x + y^2 = 0$$

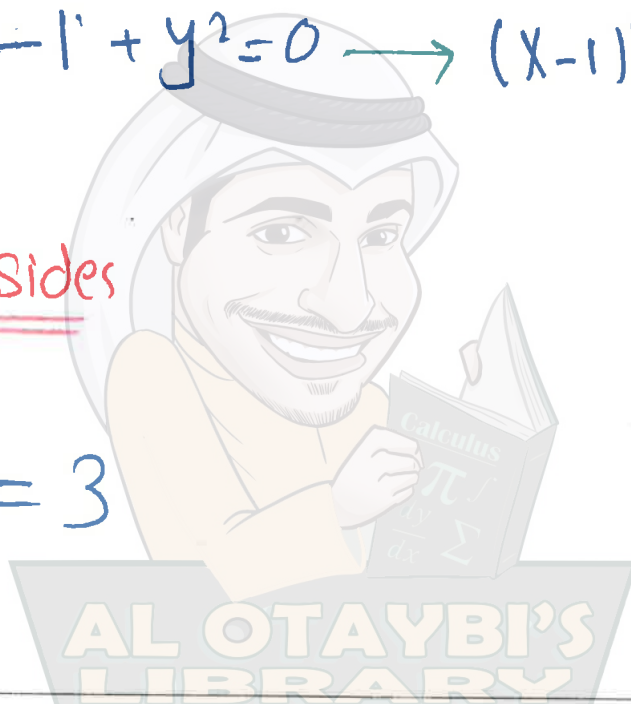
$$(x-1)^2 - 1 + y^2 = 0 \rightarrow (x-1)^2 + y^2 = 1$$

$$r = \sqrt{3}$$

Square both sides

$$r^2 = 3$$

$$x^2 + y^2 = 3$$



Let $C_1: r = 2\cos\theta$ and $C_2: r = \sqrt{3}$ be the polar curves

Find the area inside C_1 and outside C_2 .

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$$r = 2\cos\theta \quad r = \sqrt{3}$$

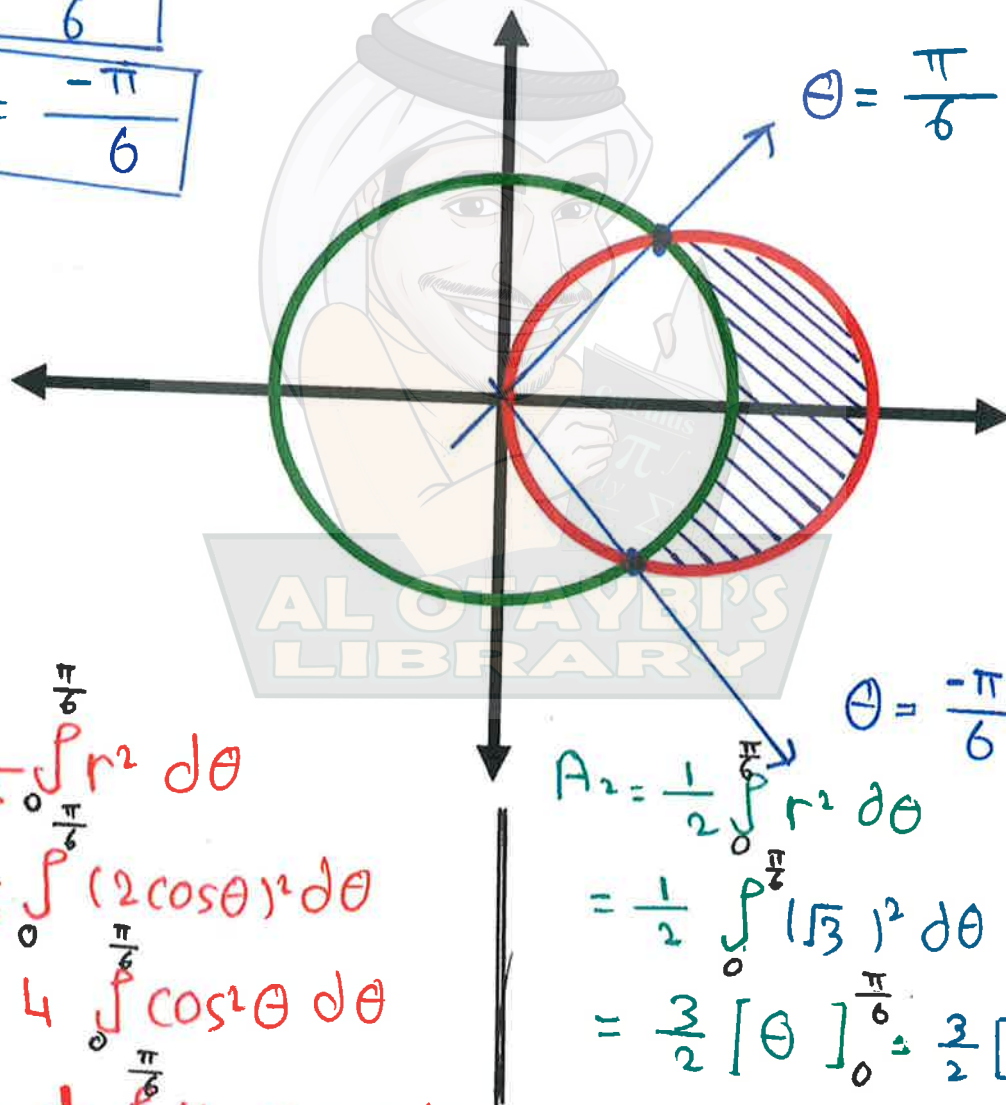
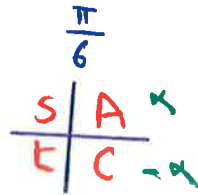
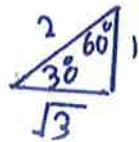
$$r = r$$

$$2\cos\theta = \sqrt{3}$$

$$\cos\theta = \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\pi}{6}$$

$$\theta = -\frac{\pi}{6}$$



$$\begin{aligned} A_1 &= \frac{1}{2} \int_0^{\pi/6} r^2 d\theta \\ &= \frac{1}{2} \int_0^{\pi/6} (2\cos\theta)^2 d\theta \\ &= \frac{1}{2} \cdot 4 \int_0^{\pi/6} \cos^2\theta d\theta \\ &= 2 \int_0^{\pi/6} (1 + \cos 2\theta) d\theta \\ &= \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/6} = \frac{\pi}{6} + \frac{\sqrt{3}}{2} - 0 \end{aligned}$$

$$\begin{aligned} A_2 &= \frac{1}{2} \int_0^{\pi/6} r^2 d\theta \\ &= \frac{1}{2} \int_0^{\pi/6} (\sqrt{3})^2 d\theta \\ &= \frac{3}{2} \left[\theta \right]_0^{\pi/6} = \frac{3}{2} \left[\frac{\pi}{6} - 0 \right] \end{aligned}$$

$$A_T = 2(A_1 - A_2) = 2\left(\frac{\pi}{6} + \frac{\sqrt{3}}{2} - \frac{3}{2} \cdot \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

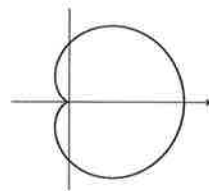
Math 102: Calculus II

Final Exam • 25 May 2017 • Duration: 120 Minutes

In questions 1-7, select the correct answer. No justification is required.

1. [4 pts] Select the polar equation that matches the given plot.

- (A) $r = 1 - \sin \theta$ (B) $r = 1 + \sin \theta$
(C) $r = 1 - \cos \theta$ (D) $r = 1 + \cos \theta$

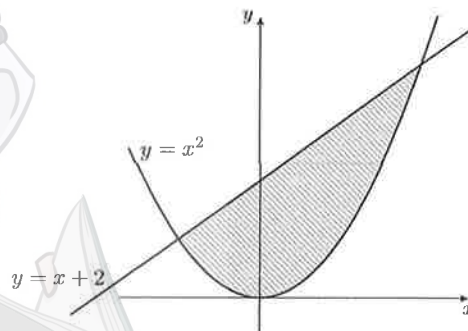


2. [4 pts] Select the integral that represents the length of the curve $y = \sin x$, $0 \leq x \leq \pi/2$.

- (A) $\int_0^1 \sqrt{1 + \cos^2 x} \, dx$ (B) $\int_0^{\pi/2} \sqrt{1 + \cos^2 x} \, dx$
(C) $\int_0^{\pi/2} 2\pi x \sqrt{1 + \cos^2 x} \, dx$ (D) $\int_0^1 \sqrt{1 + \sin^2 x} \, dx$

3. [4 pts] The shaded region is rotated about the line $y = -1$. Select the integral that represents the volume of the resulting solid.

- (A) $\int_{-1}^2 \pi \left[(x+3)^2 - (x^2+1)^2 \right] dx$
(B) $\int_{-1}^2 \pi \left[(x^2+1)^2 - (x+3)^2 \right] dx$
(C) $\int_{-2}^1 \pi \left[(x+1)^2 - (x^2-1)^2 \right] dx$
(D) $\int_{-2}^1 \pi \left[(x+3)^2 - (x^2+1)^2 \right] dx$



4. [4 pts] The partial fraction decomposition of $\frac{x^3}{x^3 + 2x^2}$ is _____.

- (A) $\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+2}$ (B) $\frac{A}{x} + \frac{Bx+C}{x^2+2}$ (C) $1 + \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+2}$ (D) $1 + \frac{A}{x} + \frac{Bx+C}{x^2+2}$

5. [4 pts] If the Maclaurin series for $f(x)$ is $\sum_{n=0}^{\infty} \frac{n^2+1}{n!} x^n$, then $f^{(5)}(0) =$ _____.

- (A) 1 (B) 5 (C) 26 (D) $\frac{26}{120}$

6. [4 pts] Let $f(x)$ be the sum of the series $\sum_{n=0}^{\infty} c_n(x+1)^n$ with radius of convergence 2. Which of the following exists?

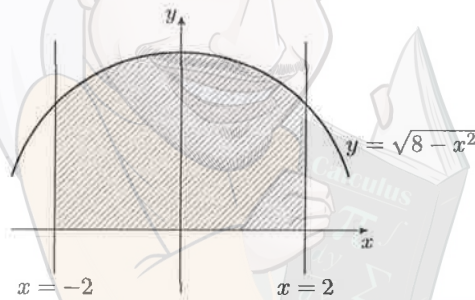
- (A) $f'(3)$ (B) $f'(-4)$ (C) $f'(2)$ (D) $f'(-2)$

7. [4 pts] The sum of the series $\sum_{n=0}^{\infty} \frac{(-4)^n \pi^{2n}}{(2n)!}$ is _____.

- (A) -1 (B) 1 (C) 0 (D) 2π

In questions 8-12, show all your work.

8. [15 pts] Determine whether the integral $\int_{-\infty}^2 x e^x dx$ is convergent or divergent. If it is convergent, find its value.
9. (a) [8 pts] Find a power series representation for $\int \ln(1 + x^3) dx$.
(b) [7 pts] Find interval of convergence of this series.
10. Let C be the curve with parametric equations $x = 2t^3$, $y = 3t^2 - 5$.
(a) [8 pts] Find an equation for the tangent line to C at the point $(16, 7)$.
(b) [7 pts] Find the length of the arc of C corresponding to $0 \leq t \leq \sqrt{3}$.
11. Let C be the polar curve $r = \sin^3(\theta/3)$, $0 \leq \theta \leq \pi$.
(a) [8 pts] Find the length of C .
(b) [7 pts] Find an equation for the tangent line to C at $\theta = \pi/2$.
12. [12 pts] Find the centroid of the shaded region.



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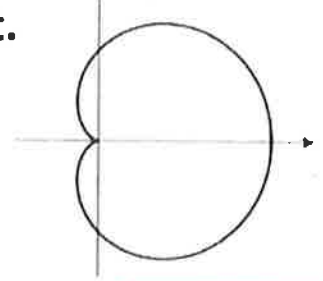
Multiple choice



Select the polar equation that matches the given plot.

(A) $r = 1 - \sin\theta$ (B) $r = 1 + \sin\theta$

(C) $r = 1 - \cos\theta$ (D) $r = 1 + \cos\theta$



May 2017

(D) ✓





Multiple choice



Select the integral that represents the length of the curve

$$y = \sin x, 0 \leq x \leq \pi/2$$

(A) $\int_0^1 \sqrt{1 + \cos^2 x} dx$

(b) $\int_0^{\pi/2} \sqrt{1 + \cos^2 x} dx$

(c) $\int_0^{\pi/2} 2\pi x \sqrt{1 + \cos^2 x} dx$

(d) $\int_0^1 \sqrt{1 + \sin^2 x} dx$

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$$L = \int_0^{\pi/2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\frac{dy}{dx} = \cos x$$

$$\left(\frac{dy}{dx}\right)^2 = \cos^2 x$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \cos^2 x$$

$$L = \int_0^{\pi/2} \sqrt{1 + \cos^2 x} dx$$

(b)



Multiple choice



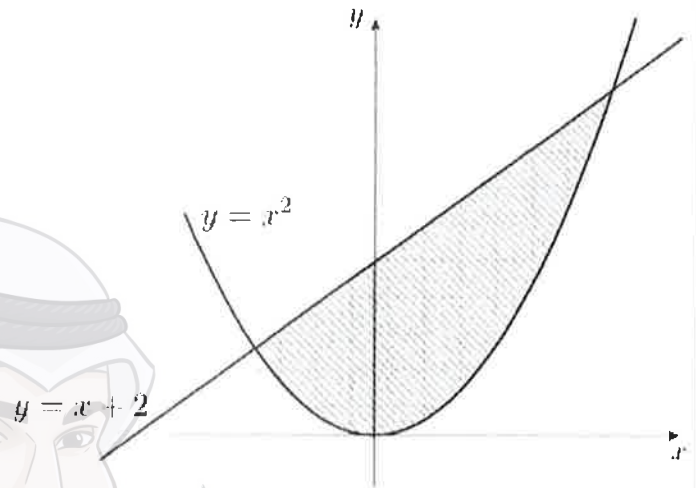
The shaded region is rotated about the line $y = -1$. Sketch the integral that represents the volume of the resulting solid.

(A) $\int_{-1}^2 \pi [(x+3)^2 - (x^2+1)^2] dx$

(b) $\int_{-1}^2 \pi [(x^2+1)^2 - (x+3)^2] dx$

(c) $\int_{-2}^1 \pi [(x+1)^2 - (x^2-1)^2] dx$

(d) $\int_{-2}^1 \pi [(x+3)^2 - (x^2+1)^2] dx$



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$y = x^2$ $y = x+2$ about $y = -1$

$y = y$

$x^2 = x+2$

$x^2 - x - 2 = 0$

$(x+1)(x-2) = 0$

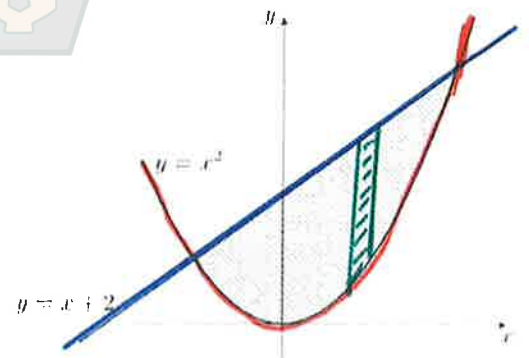
$x = -1$ $x = 2$

$y = x^2$

$y = x+2$

x	-1	0	2
y	1	0	4

x	-1	0	2
y	1	2	4



$y = -1$

$$V = \pi \int_{-1}^2 [(OR)^2 - (IR)^2] dx$$

$$= \pi \int_{-1}^2 [(x+2-(-1))^2 - (x^2-(-1))^2] dx$$

(4) The partial decomposition of

$$\frac{x^3}{x^3 + 2x^2}$$

$$\begin{array}{r} 1 \\ 2x^2 + x^3 \overline{) x^3} \\ \underline{\ominus x^3 + 2x^2} \\ -2x^2 \end{array}$$

$$f(x) = 1 + \frac{-2x^2}{x^3 + 2x^2}$$

$$\frac{-2x^2}{x^2(x+2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+2}$$

$$\Rightarrow f(x) = 1 + \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+2}$$

$\Rightarrow C$

(5) $\sum_{n=0}^{\infty} \frac{n^2+1}{n!} x^n$ $f^{(5)}(0)$?

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$\frac{n^2+1}{n!} = \frac{f^{(n)}(0)}{n!}$$

$n=5$

$$\frac{25+1}{5!} = \frac{f^{(5)}(0)}{5!}$$

*5!

$$26 = f^{(5)}(0)$$

$\Rightarrow C$

(6) $\Rightarrow D$

(7)

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$\sum_{n=0}^{\infty} \frac{(-4)^n \pi^{2n}}{(2n)!}$$

$$\sum_{n=0}^{\infty} (-1)^n \frac{4^n \pi^{2n}}{(2n)!}$$

at $x = 2\pi$

$$\cos 2\pi = \sum_{n=0}^{\infty} (-1)^n \frac{(2\pi)^{2n}}{(2n)!}$$

$$1 = \sum_{n=0}^{\infty} \frac{(-4)^n \pi^{2n}}{(2n)!}$$

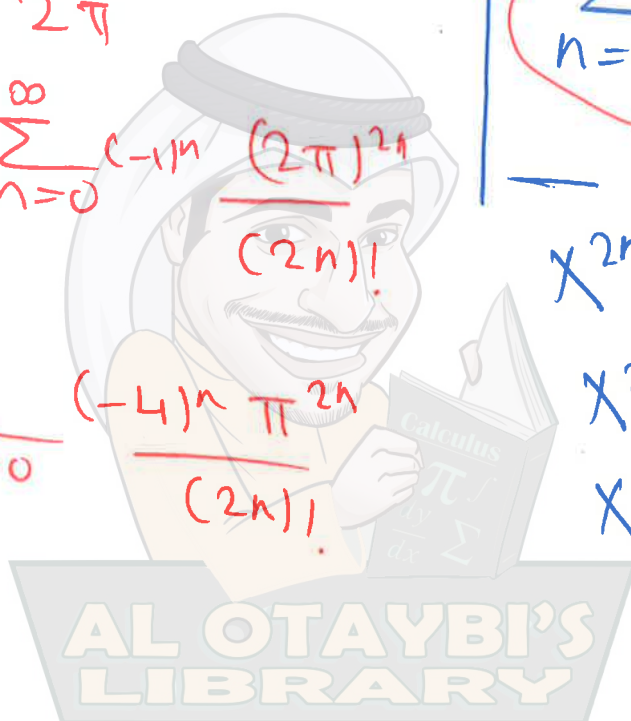
$$x^{2n} = 4^n \pi^{2n}$$

$$x^{2n} = 2^{2n} \pi^{2n}$$

$$x^{2n} = (2\pi)^{2n}$$

$$x = 2\pi$$

B



Q #8 $\int_{-\infty}^2 x e^x dx$

Conv.-Div ??

$$I_1 = \int x e^x dx$$

$$u = x \quad dv = e^x dx$$

$$du = dx \quad v = e^x$$

$$x e^x - \int e^x dx$$

$$x e^x - e^x + C$$

$$\lim_{t \rightarrow -\infty} \int_t^2 x e^x dx$$

$$\lim_{t \rightarrow -\infty} [x e^x - e^x]_t^2$$

$$\lim_{t \rightarrow -\infty} [(2e^2 - e^2) - (t e^t - e^t)]$$

$$= e^2 - (0 - 0)$$

$$= e^2$$

$\therefore$ Convergent.

$$\lim_{t \rightarrow \infty} \frac{t}{\frac{1}{e^t}} \quad \frac{\infty}{0}$$

$$\lim_{t \rightarrow -\infty} \frac{t}{e^{-t}}$$

$$\text{L.H.} \quad \lim_{t \rightarrow -\infty} \frac{1}{-e^{-t}}$$

$$= \frac{1}{-e^{\infty}} = \frac{1}{-\infty} = 0$$

- (9) (a) Find a power series for $\int \ln(1+x^3) dx$
(b) Find interval of convergence.

$$(a) \int \ln(1+x^3) dx = \int \sum_{n=0}^{\infty} (-1)^n \frac{(x^3)^{n+1}}{n+1} dx$$

$$\int \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+3}}{n+1} dx$$

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+4}}{(3n+4)(n+1)} + C$$

$$\int_{x=0}^{\infty} \ln(1+x^3) dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+4}}{(3n+4)(n+1)} + C$$

$$\underline{x=0}$$

$$0 = 0 + C \rightarrow C=0$$

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+4}}{(3n+4)(n+1)}$$

$$|x| < 1$$

$$-1 < x < 1$$

$$R = \frac{1 - (-1)}{2} = \frac{2}{2} = 1$$

at $x = -1$

$$\sum_{n=0}^{\infty} \frac{(-1)^n (-1)^{3n+4}}{(3n+4)(n+1)}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^{4n+4}}{(3n+4)(n+1)} \rightarrow (-1)^{4n} \cdot (-1)^4$$

(1) (1)

$$= \sum_{n=0}^{\infty} \frac{1}{(3n+4)(n+1)}$$

$$b_n = \frac{1}{n \cdot n} = \frac{1}{n^2}$$

$$\sum_{n=0}^{\infty} \frac{1}{n^2}$$

$p = 2 > 1$
convergent (p-series)

$$\lim_{n \rightarrow \infty} \frac{1}{(3n+4)(n+1)} \quad * n^2$$

$$\lim_{n \rightarrow \infty} \frac{1}{n(3 + \frac{4}{n})(1 + \frac{1}{n})} \quad * \frac{1}{n^2} = \frac{1}{(3+0)(1+0)} = \frac{1}{3} > 0$$

$$\therefore \sum_{n=0}^{\infty} \frac{1}{(3n+4)(n+1)} \text{ is convergent (L.C.T.)}$$

at $x = 1$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(3n+4)(n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{1}{(3n+4)(n+1)} \right| = \frac{1}{\infty} = 0$$

$$f(x) = \frac{1}{(3x+4)(x+1)}$$

$$f'(x) = \frac{0 - [3(x+1) + (3x+4)(1)]}{[(3x+4)(x+1)]^2} = \frac{-(6x+7)}{[(3x+4)(x+1)]^2}$$

$$\therefore \sum_{n=0}^{\infty} \frac{(-1)^n}{(3n+4)(n+1)} \text{ is conv. (A.S.T.)} \quad \nabla 0$$

$\Rightarrow$ interval of conv. $[-1, 1]$

$$(10) \quad x = 2t^3 \quad y = 3t^2 - 5$$

(a) Equation for tangent line at (16, 7)

$$x = 2t^3$$

$$16 = 2t^3$$

$$t^3 = 8$$

$$t = 2$$

$$y = 3t^2 - 5$$

$$7 = 3t^2 - 5$$

$$12 = 3t^2$$

$$t^2 = 4$$

$$t = \pm 2$$

$$\Rightarrow \therefore t = 2$$

$$\textcircled{1} \quad \frac{dx}{dt} = 6t^2 \quad \rightarrow \text{at } t=2 \Rightarrow \frac{dx}{dt} = 6(4) = 24$$

$$\textcircled{2} \quad \frac{dy}{dt} = 6t \quad \rightarrow \text{at } t=2 \Rightarrow \frac{dy}{dt} = 6(2) = 12$$

$$\textcircled{3} \quad \frac{dy}{dx} = \frac{12}{24} = \frac{2}{4} = \frac{1}{2}$$

$\therefore$ Equation of the tangent line

$$y - y_0 = m(x - x_0)$$

$$y - 7 = \frac{1}{2}(x - 16)$$

$\Rightarrow$ Cont.

$$(b) \quad x = 2t^3 \quad y = 3t^2 - 5$$

Find the Length. $0 \leq t \leq \sqrt{3}$

$$L = \int_0^{\sqrt{3}} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = 6t^2 \rightarrow \left(\frac{dx}{dt}\right)^2 = (6t^2)^2 = 36t^4$$

$$\frac{dy}{dt} = 6t \rightarrow \left(\frac{dy}{dt}\right)^2 = (6t)^2 = 36t^2$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 36t^4 + 36t^2$$

$$= 36t^2(t^2 + 1)$$

$$L = \int_0^{\sqrt{3}} \sqrt{36t^2(t^2 + 1)} dt$$

$$= 6 \int_0^{\sqrt{3}} t \sqrt{t^2 + 1} dt$$

$$u = t^2 + 1$$

$$du = 2t dt \rightarrow \frac{1}{2} du = t dt$$

$$= 6 \cdot \frac{1}{2} \int \sqrt{u} du = 3 \cdot \frac{2}{3} u^{\frac{3}{2}}$$

$$= 2 \left[(t^2 + 1)^{\frac{3}{2}} \right]_0^{\sqrt{3}}$$

$$= 2 \left[(3+1)^{\frac{3}{2}} - (1)^{\frac{3}{2}} \right]$$

$$= 2 \left[(2^2)^{\frac{3}{2}} - 1 \right] = 2(7)$$

$$= 14$$

Q#11 $r = \sin^3\left(\frac{\theta}{3}\right)$ $0 \leq \theta \leq \pi$

(a) Find the arc Length.

$$L = \int_0^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$r = \sin^3\left(\frac{\theta}{3}\right) \rightarrow r^2 = \sin^6\left(\frac{\theta}{3}\right)$$

$$\frac{dr}{d\theta} = 3 \sin^2 \frac{\theta}{3} \cos \frac{\theta}{3} \cdot \frac{1}{3}$$

$$= \sin^2 \frac{\theta}{3} \cos \frac{\theta}{3}$$

$$\rightarrow \left(\frac{dr}{d\theta}\right)^2 = \sin^4 \frac{\theta}{3} \cos^2 \frac{\theta}{3}$$

$$\begin{aligned} \Rightarrow r^2 + \left(\frac{dr}{d\theta}\right)^2 &= \sin^6\left(\frac{\theta}{3}\right) + \sin^4 \frac{\theta}{3} \cos^2 \frac{\theta}{3} \\ &= \sin^4\left(\frac{\theta}{3}\right) \left[\sin^2 \frac{\theta}{3} + \cos^2 \frac{\theta}{3} \right] \\ &= \sin^4\left(\frac{\theta}{3}\right) \quad (1) \end{aligned}$$

$$L = \int_0^{\pi} \sqrt{\sin^4\left(\frac{\theta}{3}\right)} d\theta$$

$$= \int_0^{\pi} \sin^2 \frac{\theta}{3} d\theta$$

$$= \frac{1}{2} \int_0^{\pi} [1 - \cos \frac{2\theta}{3}] d\theta$$

$$= \frac{1}{2} \left[\theta - \frac{3}{2} \sin \frac{2\theta}{3} \right]_0^{\pi}$$

$$= \frac{1}{2} \left[\pi - \frac{3}{2} \cdot 0 - 0 \right]$$

$$(b) \quad r = \sin^3\left(\frac{\theta}{3}\right)$$

Find equation of tangent line at $\theta = \frac{\pi}{2}$

$$x = r \cos \theta = \sin^3 \frac{\theta}{3} \cos \theta$$

$$y = r \sin \theta = \sin^3 \frac{\theta}{3} \sin \theta$$

$$\text{at } \theta = \frac{\pi}{2} \quad \begin{cases} x = 0 \\ y = \sin^3 \frac{\pi}{6} (1) = \left(\frac{1}{2}\right)^3 = \frac{1}{8} \end{cases} \Rightarrow (0, \frac{1}{8})$$

$$\begin{aligned} \frac{dx}{d\theta} &= 3 \sin^2 \frac{\theta}{3} \cos \frac{\theta}{3} \cdot \frac{1}{3} \cos \theta + \sin^3 \frac{\theta}{3} (-\sin \theta) \\ &= \sin^2 \frac{\theta}{3} \cos \frac{\theta}{3} \cos \theta - \sin^3 \frac{\theta}{3} \sin \theta \end{aligned}$$

$$\text{at } \theta = \frac{\pi}{2} \rightarrow \frac{dx}{d\theta} = \left(\frac{1}{2}\right)^2 \frac{\sqrt{3}}{2} \cdot 0 - \left(\frac{1}{2}\right)^3 (1) = -\frac{1}{8}$$

$$\frac{dy}{d\theta} = 3 \sin^2 \frac{\theta}{3} \cos \frac{\theta}{3} \cdot \frac{1}{3} \sin \theta + \sin^3 \frac{\theta}{3} \cos \theta$$

$$\begin{aligned} \text{at } \theta = \frac{\pi}{2} \rightarrow \frac{dy}{d\theta} &= 3 \left(\frac{1}{2}\right)^2 \frac{\sqrt{3}}{2} \cdot \frac{1}{3} (1) + 0 \\ &= \frac{1}{4} \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{8} \end{aligned}$$

$$\frac{dy}{dx} = \frac{\frac{\sqrt{3}}{8}}{-\frac{1}{8}} = -\sqrt{3}$$

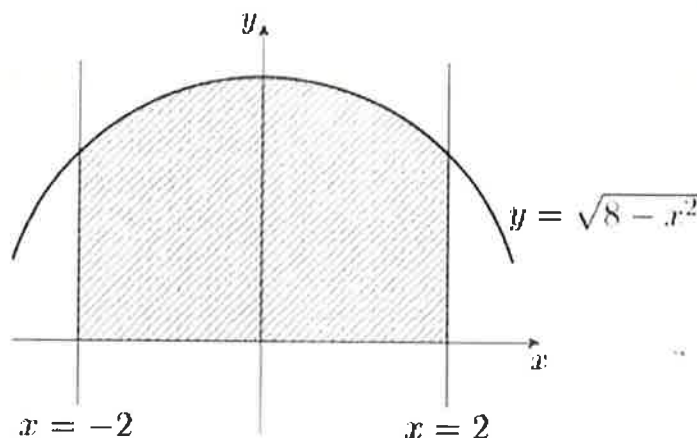
$\therefore$ Equation of tangent line

$$y - y_0 = m(x - x_0)$$

$$y - \frac{1}{8} = -\sqrt{3}(x - 0)$$

Find the centroid of the shaded region.

May 2017

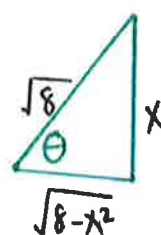


From the region $\rightarrow \bar{x} = 0$

$$A = \int_{-2}^2 [\sqrt{8-x^2} - 0] dx$$

Let $x = \sqrt{8} \sin \theta \rightarrow \sin \theta = \frac{x}{\sqrt{8}}$

$$dx = \sqrt{8} \cos \theta d\theta$$



$$= \int \sqrt{8-8\sin^2\theta} \sqrt{8} \cos \theta d\theta$$

$$\sqrt{8(1-\sin^2\theta)} = \sqrt{8\cos^2\theta} = \sqrt{8} \cos \theta$$

$$= \int \sqrt{8} \cos \theta \sqrt{8} \cos \theta d\theta$$

$$= 8 \int \cos^2 \theta d\theta = 8 \cdot \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$= 4 \left[\theta + \frac{\sin 2\theta}{2} \right] = 4 \left[\theta + \frac{2\sin \theta \cos \theta}{2} \right]$$

$$= 4 \left[\sin^{-1} \left(\frac{x}{\sqrt{8}} \right) + \frac{x}{\sqrt{8}} \cdot \frac{\sqrt{8-x^2}}{\sqrt{8}} \right]_{-2}^2$$

$$\sqrt{8} = \sqrt{4(2)} = 2\sqrt{2}$$

$$= 4 \left[\left(\sin^{-1} \left(\frac{1}{\sqrt{2}} \right) + \frac{1}{\sqrt{2}} \cdot \frac{2}{2\sqrt{2}} \right) - \left(\sin^{-1} \left(\frac{-1}{\sqrt{2}} \right) - \frac{1}{\sqrt{2}} \cdot \frac{2}{2\sqrt{2}} \right) \right]$$

$$= 4 \left[\frac{\pi}{4} + \frac{1}{2} + \frac{\pi}{4} + \frac{1}{2} \right] = 4 \left[\frac{\pi}{2} + 1 \right] = 2\pi + 4$$

$$\bar{y} = \frac{1}{2A} \int_{-2}^2 [(\sqrt{8-x^2})^2 - 0^2] dx$$

$$= \frac{1}{2[2\pi+4]} \int_{-2}^2 [8-x^2] dx$$

$$= \frac{1}{2[2\pi+4]} \left[8x - \frac{x^3}{3} \right]_{-2}^2$$

$$= \frac{1}{2[2\pi+4]} \left[16 - \frac{8}{3} - \left(-16 + \frac{8}{3} \right) \right]$$

$$= \frac{1}{2[2\pi+4]} \left[32 - \frac{16}{3} \right]$$

$$= \frac{1}{2(2)[\pi+2]} \frac{80}{3} = \frac{20}{3\pi+6}$$

$$\Rightarrow \therefore P(\bar{x}, y) = P\left(0, \frac{20}{3\pi+6}\right)$$